

A Hybrid Classical–Learning Framework for Adaptive Decision-Directed Speech Enhancement

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Abstract—Speech enhancement aims to recover clean speech signals from noisy observations while preserving speech quality and intelligibility. Classical methods such as Spectral Subtraction and Decision-Directed (DD) enhancement remain widely used because of their interpretability and low computational complexity, but they may suffer from musical-noise artifacts or excessive attenuation of weak speech components under low signal-to-noise ratio (SNR) conditions. This paper proposes an Adaptive Beta-Constrained Decision-Directed (ABCDD) speech enhancement framework that extends the conventional DD method through a frame-dependent lower gain bound. The introduced beta parameter controls the tradeoff between noise suppression and speech preservation. To automate parameter selection for large and diverse datasets, a lightweight multilayer perceptron (MLP) model is further developed to predict frame-level beta values directly from noisy-speech features. The proposed framework is evaluated using both a representative speech example and large-scale testing on the VoiceBank-DEMAND dataset. In the representative example, ABCDD outperformed conventional Spectral Subtraction and classical DD across multiple objective metrics, including SNR, Log-Spectral Distance (LSD), Root-Mean-Square Error (RMSE), correlation, and Scale-Invariant Signal-to-Distortion Ratio (SI-SDR). On 100 unseen VoiceBank-DEMAND test files, the proposed MLP-beta ABCDD method improved average scale-aligned SNR from 9.41 dB to 13.82 dB, corresponding to an average gain of 4.41 dB. The results indicate that combining interpretable classical enhancement structure with lightweight machine-learning-based parameter adaptation provides an effective and practical direction for robust speech enhancement.

Index Terms—Speech Enhancement, Decision-Directed Estimation, Spectral Subtraction, Machine Learning, Adaptive Filtering.

I. INTRODUCTION

Speech enhancement aims to recover a clean speech signal from observations degraded by additive background noise. The problem remains challenging because practical acoustic noise is often time-varying and non-stationary. Reliable speech enhancement is important in communication systems, hearing devices, speech recording, teleconferencing, and robust automatic speech processing applications [1], [2].

Because both speech and noise characteristics vary across time and frequency, many enhancement methods operate in the

short-time spectral domain, where noisy speech is processed frame by frame after time-frequency transformation. Early approaches in this category include Spectral Subtraction, one of the earliest and most widely used speech enhancement methods [3]. Later refinements introduced over-subtraction and spectral flooring to reduce residual noise and improve robustness [4]. These methods are widely used because of their conceptual simplicity and low computational complexity. However, Spectral Subtraction is sensitive to noise-estimation errors and may introduce speech distortion or musical-noise artifacts [3], [4].

To address these limitations, gain-based statistical enhancement methods such as Wiener filtering and minimum mean-square error (MMSE) estimators were developed [5], [6]. Decision-directed (DD) enhancement further improved temporal smoothness by recursively estimating the *a priori* signal-to-noise ratio (SNR) from previous enhanced frames and current observations [7]. Nevertheless, DD gains may become excessively small in low-SNR regions, which can lead to over-attenuation of weak speech components.

These limitations highlight the need for enhancement methods that suppress noise effectively while adaptively preserving weak speech content. Accurate noise estimation is also important in practical systems. Minimum-statistics and recursive noise-tracking methods were developed to estimate non-stationary noise power spectral density (PSD) under realistic acoustic conditions [8]. Other approaches incorporated speech-presence uncertainty to improve robustness in changing environments [9]. Spectral-flooring strategies have also remained useful in modern real-time systems because they help reduce over-suppression and limit artifacts [10].

Broader prior studies in speech enhancement can be grouped into three main directions. First, classical spectral-domain approaches focused on subtraction rules and statistical gain estimation, including Spectral Subtraction [3], Wiener filtering [5], MMSE estimators [6], and DD-based enhancement [7]. Second, adaptive noise-estimation techniques improved robustness under non-stationary acoustic environments [8], [9]. Third, recent learning-based methods have learned direct

mappings from noisy speech to clean speech representations using feedforward neural networks [2], recurrent neural networks [11], convolutional time-domain models [12], and lightweight real-time architectures [1]. Deep learning has also been used to improve classical enhancement components such as noise PSD estimation [13]. Hybrid approaches combining signal processing and learning-based adaptation have likewise shown practical benefits [14].

Although modern neural approaches can achieve strong enhancement performance, many purely data-driven models reduce interpretability and may require greater computational resources than classical gain-based methods. On the other hand, conventional DD enhancement preserves a transparent and efficient structure, but its gain may over-suppress weak speech when local SNR becomes low. This motivates adaptive frameworks that retain classical interpretability while introducing data-driven parameter control.

In this work, an Adaptive Beta-Constrained Decision-Directed (ABCDD) enhancement framework is proposed. The method modifies the conventional DD gain by imposing a frame-dependent lower gain bound controlled by a beta parameter. This beta value governs the tradeoff between noise suppression and speech preservation. Smaller beta values allow stronger suppression but may distort weak speech, whereas larger beta values preserve speech components but may leave residual noise.

Manual beta tuning may be feasible for a single speech example, but it becomes impractical for large datasets containing many speakers, utterances, and noise conditions. Therefore, this paper further proposes a lightweight multilayer perceptron (MLP)-based beta prediction strategy for ABCDD enhancement. Instead of manually selecting beta values, the trained MLP predicts frame-dependent beta values from noisy-speech features. The model is trained and evaluated using the VoiceBank-DEMAND dataset, which provides paired clean and noisy speech examples for speech enhancement experiments [11].

The main contributions of this work are summarized as follows. (1) An ABCDD enhancement framework is developed to improve the tradeoff between speech preservation and noise suppression. (2) A lightweight MLP model is developed to predict frame-level beta values automatically from noisy-speech features. (3) The proposed method is validated on a representative speech example and on 100 unseen VoiceBank-DEMAND test files, demonstrating improved enhancement performance.

II. BACKGROUND AND PRELIMINARIES

This section reviews the fundamental signal-processing methods that motivate the proposed framework. Since the objective of speech enhancement is to estimate clean speech from noisy observations, classical spectral-domain approaches such as spectral subtraction and decision-directed (DD) estimation provide useful baselines. These methods also form

the foundation of the constrained decision-directed (CDD) enhancement strategy developed later in this paper.

A. Noisy Speech Model and STFT Representation

The observed noisy speech signal is commonly modeled as the sum of clean speech and additive noise:

$$x(n) = s(n) + v(n) \quad (1)$$

where $x(n)$, $s(n)$, and $v(n)$ denote the noisy speech, clean speech, and noise signals, respectively. This additive model is widely used in speech enhancement studies [3].

Because speech and environmental noise are nonstationary, their short-term energy and spectral content vary over time. Therefore, enhancement is typically performed in the short-time Fourier transform (STFT) domain [5]. The STFT of $x(n)$ is defined as

$$X(k, m) = \sum_{r=0}^{L-1} x(r + mR)w(r)e^{-j2\pi kr/N} \quad (2)$$

where k is the frequency-bin index, m is the frame index, $w(r)$ is the analysis window, L is the frame length, R is the hop size, and N is the FFT size.

Under the additive model, the noisy spectral coefficient can be written as

$$X(k, m) = S(k, m) + V(k, m) \quad (3)$$

where $S(k, m)$ and $V(k, m)$ represent the clean speech and noise spectral components, respectively.

This time-frequency representation enables spectral subtraction to attenuate estimated noise bins [3], Wiener filtering to perform statistical gain control [5], and decision-directed methods to recursively smooth spectral estimates [6].

After enhancement, the modified spectrum $\hat{S}(k, m)$ is converted back to the time domain using the inverse STFT:

$$\hat{s}_m(r) = \frac{1}{N} \sum_{k=0}^{N-1} \hat{S}(k, m)e^{j2\pi kr/N} \quad (4)$$

The reconstructed short-time frames are then shifted and summed using overlap-add synthesis to produce the final enhanced waveform $\hat{s}(n)$.

B. Spectral Subtraction Method

Spectral subtraction is one of the earliest and most widely used speech enhancement techniques. The method estimates clean speech by subtracting an estimate of the noise spectrum from the noisy speech spectrum in the STFT domain [3]. Let the noisy spectrum satisfy

$$X(k, m) = S(k, m) + V(k, m) \quad (5)$$

where $X(k, m)$, $S(k, m)$, and $V(k, m)$ denote the noisy speech, clean speech, and noise spectral coefficients, respectively.

In practice, the noise magnitude spectrum $|\hat{V}(k, m)|$ is commonly obtained from noise-only segments, low-energy

frames, or recursive noise power estimation methods [8]. Using this estimate, the enhanced speech magnitude can be written as

$$|\hat{S}(k, m)| = \max(|X(k, m)| - |\hat{V}(k, m)|, 0) \quad (6)$$

where the nonnegative constraint avoids invalid negative magnitudes after subtraction.

To improve robustness, an over-subtraction factor α_s and a spectral floor parameter β_s are often introduced [4]:

$$|\hat{S}(k, m)| = \max(|X(k, m)| - \alpha_s |\hat{V}(k, m)|, \beta_s |X(k, m)|) \quad (7)$$

where $\alpha_s > 1$ increases suppression strength and β_s prevents excessive attenuation.

Because only the magnitude spectrum is modified, the noisy phase is typically preserved:

$$\hat{S}(k, m) = |\hat{S}(k, m)| e^{j\angle X(k, m)} \quad (8)$$

The enhanced time-domain signal is then recovered through the inverse STFT and overlap-add synthesis in (4).

Spectral subtraction is still useful because it is computationally efficient and straightforward to implement. However, inaccurate noise estimation may generate residual tonal artifacts known as musical noise, while aggressive subtraction can distort weak speech components [15]. These limitations motivate more adaptive gain-based methods such as Wiener filtering and decision-directed enhancement.

C. Decision-Directed (DD) Enhancement

Decision-directed (DD) enhancement is a gain-based spectral-domain method that improves speech continuity by recursively estimating the *a priori* signal-to-noise ratio (SNR). Compared to spectral subtraction method, DD method generally reduce musical noise and provide smoother frame-to-frame enhancement behavior [6], [7].

Let $\lambda_v(k, m)$ denote the noise power spectral density (PSD). The *a posteriori* SNR is computed from the noisy observation as

$$\gamma(k, m) = \frac{|X(k, m)|^2}{\lambda_v(k, m)} \quad (9)$$

where $X(k, m)$ is the noisy speech STFT coefficient.

Instead of relying only on the current frame, the DD approach recursively estimates the *a priori* SNR using both the previous enhanced frame and the present observation:

$$\xi_{DD}(k, m) = \alpha \frac{|\hat{S}(k, m-1)|^2}{\lambda_v(k, m)} + (1-\alpha) \max(\gamma(k, m) - 1, 0) \quad (10)$$

where $\xi_{DD}(k, m)$ denotes the DD estimate of the *a priori* SNR, $\hat{S}(k, m-1)$ is the enhanced speech spectrum from the previous frame, and $0 < \alpha < 1$ is a smoothing parameter that controls the tradeoff between relying on the previous enhanced frame and the current noisy observation.

The first term preserves temporal continuity using past enhanced speech, while the second term incorporates

current-frame information. This balance helps suppress rapid gain fluctuations that often create residual artifacts.

Using the estimated *a priori* SNR, a Wiener-type gain is obtained as

$$G_{DD}(k, m) = \frac{\xi_{DD}(k, m)}{1 + \xi_{DD}(k, m)} \quad (11)$$

The enhanced spectrum is then computed by

$$\hat{S}(k, m) = G_{DD}(k, m) X(k, m) \quad (12)$$

Finally, the enhanced time-domain signal is reconstructed through the inverse STFT and overlap-add synthesis in (4).

Decision-directed enhancement is effective because it provides smoother suppression than basic spectral subtraction and usually preserves speech quality more naturally. However, when $\xi_{DD}(k, m)$ becomes very small in low-SNR regions, the gain in (11) may excessively attenuate weak speech components. This limitation motivates the constrained decision-directed method introduced next.

III. PROPOSED METHODOLOGY

This section presents the proposed enhancement framework developed in this work. Starting from the conventional decision-directed (DD) method, an adaptive beta-constrained formulation is introduced to improve the tradeoff between noise suppression and speech preservation. Motivated by the difficulty of manually selecting suitable beta values over large datasets, a machine-learning-based strategy is then developed to predict frame-dependent beta values for the proposed enhancement system.

A. Adaptive Beta-Constrained Decision-Directed (ABCDD) Enhancement

Although DD enhancement provides smoother suppression than spectral subtraction, the gain in low-SNR regions may become excessively small and attenuate weak speech components. To address this limitation, the proposed Adaptive Beta-Constrained Decision-Directed (ABCDD) method introduces a lower gain bound controlled by a frame-dependent parameter β_m , where $\beta_m \in [0, 1]$. Starting from the DD gain in (11), the constrained gain is defined as

$$G_{ABCDD}(k, m) = \max(G_{DD}(k, m), \beta_m) \quad (13)$$

where β_m is the adaptive beta parameter associated with frame m . This modification imposes a lower gain floor on the conventional DD estimator. As a result, excessive attenuation in low-SNR regions can be reduced, while a tunable balance between noise suppression and speech preservation is maintained. The enhanced spectrum is then computed as

$$\hat{S}(k, m) = G_{ABCDD}(k, m) X(k, m) \quad (14)$$

The reconstructed short-time frames are then shifted and summed using overlap-add synthesis to produce the final enhanced waveform which is reconstructed using the ISTFT and overlap-add synthesis in (4).

The parameter β_m directly controls the behavior of the ABCDD method. Smaller values allow stronger noise suppression but may distort weak speech, whereas larger values preserve speech components while reducing noise suppression strength. If β_m becomes too large, residual noise may remain audible. If β_m becomes too small, the method approaches conventional DD behavior and may over-suppress low-energy speech components. Therefore, different beta settings may lead to noticeably different output waveforms and signal-to-noise ratio (SNR) values. Thus, selecting appropriate beta values is crucial to ABCDD performance.

For a given speech signal, the beta parameter can be manually adjusted to improve enhancement quality. By varying β_m , the balance between residual noise and speech preservation can be controlled. Although manual tuning may be effective for a single speech example, the preferred beta value can vary across speakers, utterances (audio samples), and noise conditions. As a result, manual beta selection is impractical for large datasets and motivates automatic beta prediction.

B. MLP-Based Beta Prediction for ABCDD Enhancement

To automate beta selection for the proposed ABCDD method, a multilayer perceptron (MLP) model is developed to predict frame-dependent beta values directly from noisy-speech frame features. Instead of manually tuning β_m for each speech sample, the trained network estimates suitable beta values in a data-driven manner.

For each frame m , a feature vector is extracted as

$$\mathbf{f}_m = [f_1, f_2, \dots, f_P]^T \quad (15)$$

where P denotes the number of selected features that include posterior SNR statistics, DD gain statistics, frame energy, and spectral flatness. These features provide information regarding noise level, spectral structure, and enhancement behavior of the current frame.

The MLP learns a non-linear mapping between the input features and the desired beta value:

$$\beta_m = \Phi(\mathbf{f}_m; \Theta) \quad (16)$$

where $\Phi(\cdot)$ denotes the neural network model and Θ represents the trainable parameters, including layer weights and biases. The implemented MLP consists of two fully connected hidden layers with 64 and 32 neurons, respectively, followed by rectified linear unit (ReLU) activations. A single-output regression node was used to predict the beta value. L2 regularization with coefficient 10^{-4} was applied to reduce overfitting, and the network was trained for up to 300 iterations using MATLAB's built-in regression neural network optimizer. These settings were selected to provide sufficient non-linear modeling capacity while maintaining low computational complexity.

VoiceBank-DEMAND dataset was used to generate supervised training examples under diverse speakers and noise conditions [11]. Noisy speech files were processed frame

by frame, and feature vectors were extracted from each frame. Each speech recording was segmented into overlapping short-time frames using a 25 ms analysis window and 10 ms frame shift. The number of frames extracted from each file depended on signal duration, with typical recordings producing several hundred frames. A subset of 500 noisy-clean paired utterances from the VoiceBank-DEMAND training dataset was used for model development. After frame segmentation and feature extraction, these recordings generated 164,692 labeled frame-level examples.

For each training frame, candidate beta values were selected and evaluated over a finite predefined uniform search grid in $[0, 1]$. The beta value that minimized the enhancement loss with respect to the corresponding clean reference signal, such as reduced reconstruction error or improved SNR, was selected as the oracle target label. Therefore, each label represents the best beta value within the tested search range. These oracle beta labels were then paired with the extracted feature vectors to form the supervised training dataset.

The dataset was randomly divided into 80% training data (131,754 frames) and 20% validation data (32,938 frames). The training subset was used to optimize the network parameters, while the validation subset was used to evaluate generalization performance and reduce overfitting. Model training minimized mean squared error between predicted and oracle beta labels. Hyperparameters such as the number of hidden layers, number of neurons, learning iterations, regularization level, and activation type were selected based on validation performance.

After training, the final model was evaluated on 100 unseen noisy speech files. During enhancement, the trained model predicts β_m for each frame, and the predicted value is inserted into (13) to generate the adaptive enhancement gain. This data-driven approach enables scalable parameter adaptation while retaining the interpretable ABCDD enhancement structure. The proposed framework was implemented in MATLAB and evaluated using a single speech example and the VoiceBank-DEMAND dataset. The following section presents the experimental setup and comparative performance results.

IV. EXPERIMENTAL SETUP AND RESULTS

This section evaluates the proposed ABCDD framework using a representative speech example and large-scale dataset testing. First, Spectral Subtraction (SpecSub), DD, and ABCDD are compared on a controlled noisy signal. Then, the MLP-based beta prediction method is evaluated on the VoiceBank-DEMAND dataset.

A. Representative Speech Example Evaluation

Let Sig_{OG} denote the clean reference signal, and let additive noise generate the corrupted signal Sig_{Noisy} . SpecSub, DD, and ABCDD were applied to the same noisy input using identical STFT settings for fair comparison. Enhanced signals were reconstructed using the ISTFT.

Performance was assessed using waveform inspection and objective metrics relative to the clean reference. Global

TABLE I: Representative Speech Example Results for *SigNoisy*

Method	SNR (dB)	SegSNR (dB)	LSD (dB)	RMSE	Corr.	SI-SDR (dB)
Noisy	-0.94	-6.67	42.19	0.0156	0.6650	-1.01
SpecSub	-0.09	-0.03	23.08	0.0141	0.0143	-36.91
DD	11.88	4.77	19.61	0.0036	0.9710	12.17
ABCDD	12.22	5.25	15.46	0.0034	0.9739	12.65

SNR measures overall noise reduction, Segmental SNR averages frame-wise SNR, LSD measures spectral distortion, RMSE measures sample-wise error, correlation measures waveform similarity, and SI-SDR evaluates distortion after scale alignment.

Fig. 1 summarizes the representative-example results.

Table I shows that ABCDD achieved the best performance for all reported metrics. Compared to DD, ABCDD improved SNR, Segmental SNR, Log-Spectral Distance (LSD), Root-Mean-Square Error (RMSE), correlation, and Scale-Invariant Signal-to-Distortion Ratio (SI-SDR), confirming that the adaptive beta floor improves the tradeoff between noise suppression and speech preservation.

SpecSub reduced some noise energy and spectral distortion, but its low correlation and poor SI-SDR indicate weak preservation of speech structure. This behavior is consistent with known spectral-subtraction artifacts such as speech distortion and musical noise.

DD substantially outperformed SpecSub through smoother recursive gain adaptation. ABCDD further improved DD by preserving weak speech components while maintaining effective suppression.

B. Large-Scale VoiceBank-DEMAND Evaluation

To evaluate scalability, the proposed MLP-based beta prediction framework was tested on the VoiceBank-DEMAND dataset. A subset of 500 noisy-clean paired utterances was used for training, and the final model was evaluated on 100 unseen noisy speech files.

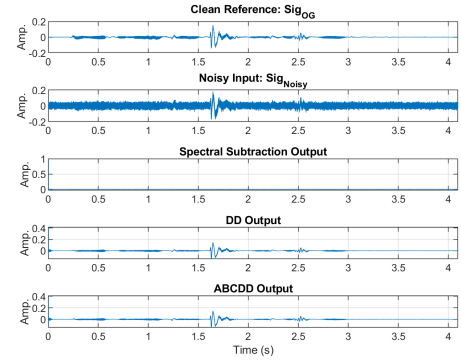
For each test signal, frame-level features were extracted and used by the trained MLP to predict adaptive beta values, which were then applied within the ABCDD gain function.

Average scale-aligned SNR was used as the primary metric. The noisy test files achieved an average SNR of 9.41 dB, while the proposed MLP-beta ABCDD method achieved 13.82 dB, corresponding to an average SNR improvement of 4.41 dB.

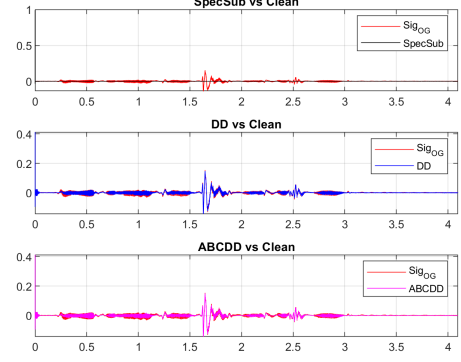
Fig. 2 presents the average SNR comparison for the 100 unseen test utterances. The positive gain indicates that the trained model learned effective beta adaptation across diverse noisy speech conditions.

V. CONCLUSION

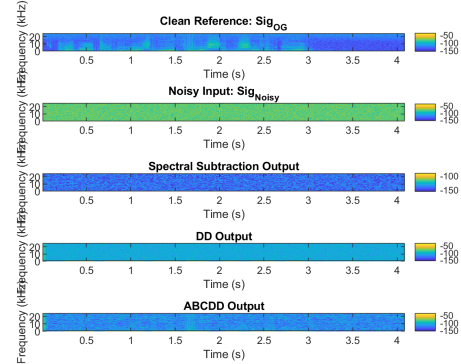
This paper presented ABCDD speech enhancement framework that extends the classical DD method through a frame-dependent lower gain bound. The proposed gain-floor control improves the balance between noise suppression and



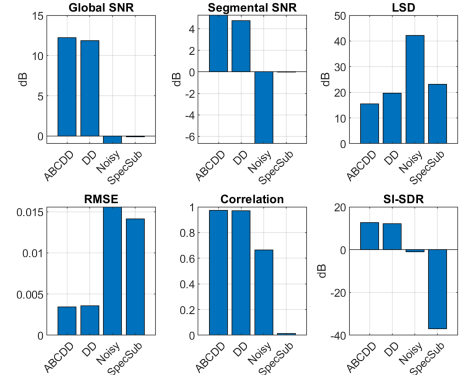
(a) Time-domain waveforms



(b) Overlay with clean reference



(c) Spectrogram comparison



(d) Objective metric comparison

Fig. 1: Representative speech example evaluation for *SigNoisy* using SpecSub, DD, and ABCDD.

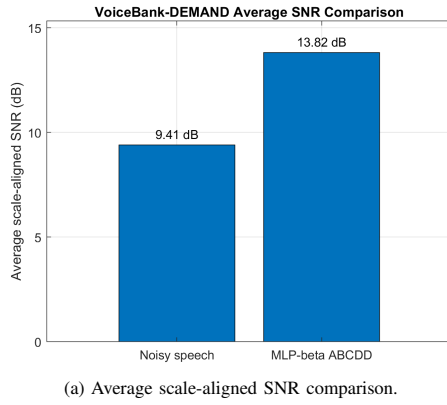


Fig. 2: Large-scale VoiceBank-DEMAND evaluation of the proposed MLP-beta ABCDD method.

speech preservation, particularly for weak speech components in low-SNR regions.

For large datasets such as VoiceBank-DEMAND, automatic parameter selection is required. Therefore, an MLP model was developed and trained on the dataset to predict frame-level beta values from noisy-speech features. This enables scalable enhancement without manual tuning for each utterance.

Experimental results on a representative speech example showed that ABCDD outperformed conventional Spectral Subtraction and classical DD across multiple objective metrics, including SNR, LSD, RMSE, correlation, and SI-SDR. Large-scale evaluation on the VoiceBank-DEMAND dataset further demonstrated that the proposed MLP-beta ABCDD method improved average scale-aligned SNR by 4.41 dB over the noisy baseline across 100 unseen test files.

Overall, the results indicate that combining interpretable classical enhancement structure with lightweight machine-learning-based parameter adaptation is a practical direction for robust speech enhancement. Future work may consider deeper neural architectures, perceptual quality metrics, real-time implementation, and evaluation under broader non-stationary noise conditions.

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